

Name: Neel Patel

Roll no: 33

Class : Btech 3

| Experiment No. – 8                                                      |                              |              |                          |                |
|-------------------------------------------------------------------------|------------------------------|--------------|--------------------------|----------------|
| <b>Date of Performance:</b>                                             | 3/10/2024                    |              |                          |                |
| <b>Date of Submission:</b>                                              | 10/10/2024                   |              |                          |                |
| Program<br>Execution/formation/<br>correction/ethical practices<br>(06) | Timely<br>Submission<br>(01) | Viva<br>(03) | Experiment<br>Total (10) | Sign with Date |
|                                                                         |                              |              |                          |                |

### Experiment No. 8

#### Timeline Report

**8.1 Aim :** To generate a comprehensive timeline report using Autopsy, enabling the reconstruction of digital events and activities.

#### **8.2 Course Outcome :**

Discuss the terminologies related to Digital forensics and computer security incident process. Describe and apply the process of collection, analysis and recovery of the digital evidence.

**8.3 Learning Objectives:** To develop skills in documenting and reporting findings from the timeline analysis to support forensic investigations and incident response.

**8.4 Requirement:** Autopsy

#### **8.5 Related Theory :**

Autopsy allows you to generate a report—in HTML, Excel, text, and other formats—that contains time

information of every file in the supplied forensic image. This feature opens possibilities to use such information in other programs outside Autopsy.

## 8.6 Procedure:

To generate a timeline report using Autopsy, follow these steps:

1. Go to Tools menu ➤ Generate Report. The Generate Report wizard appears; the first window allows you to select the report format (see Figure 8.1).

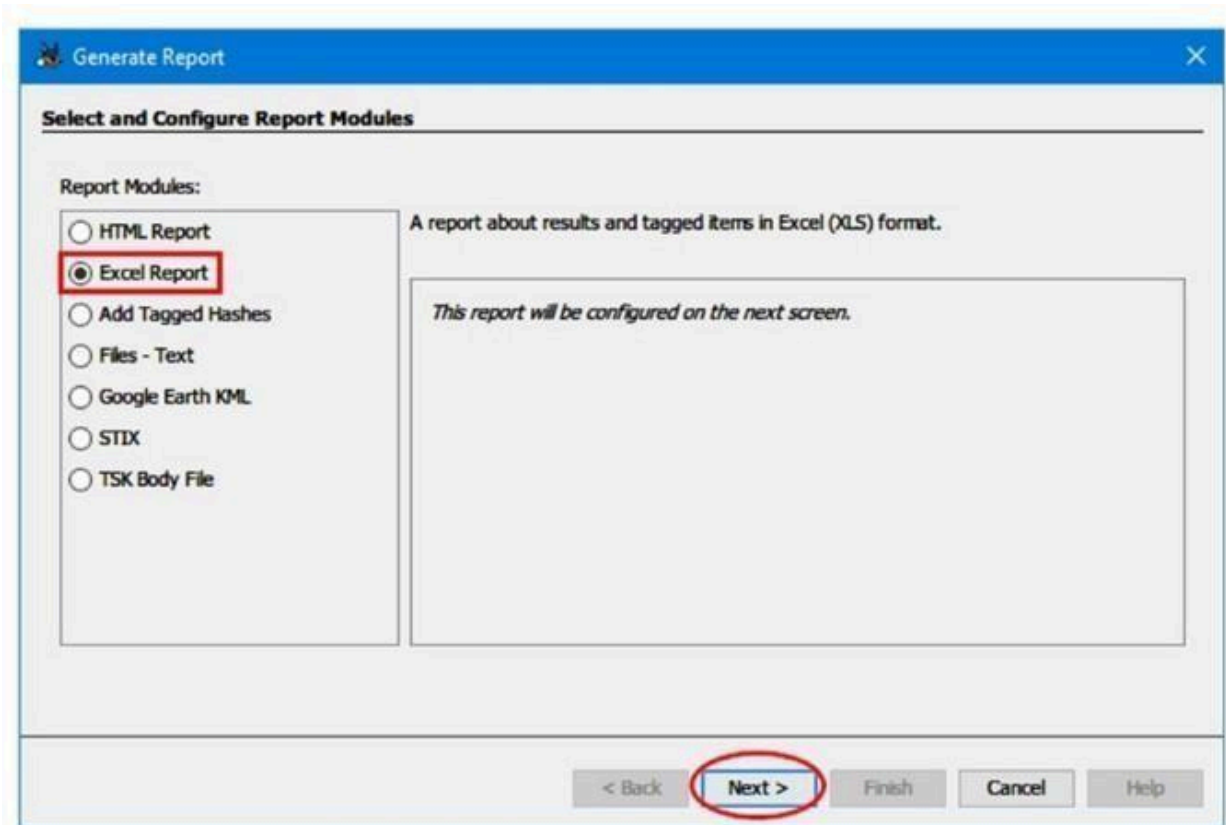


Figure 8.1. Select report format for your generated timeline in Autopsy

2. In our case, we select “Excel Report,” so we can play with the data using the MS Excel spreadsheet program or any other alternative program that can read Excel files like Apache OpenOffice ([www.openoffice.org](http://www.openoffice.org)). Click “Next” to continue.
3. The next window asks you to configure the returned results. You have two options: All Results and Tagged Results. In our case, we will select all results and click “Finish”; then, Autopsy will begin the report generation process (see Figure 8.2).

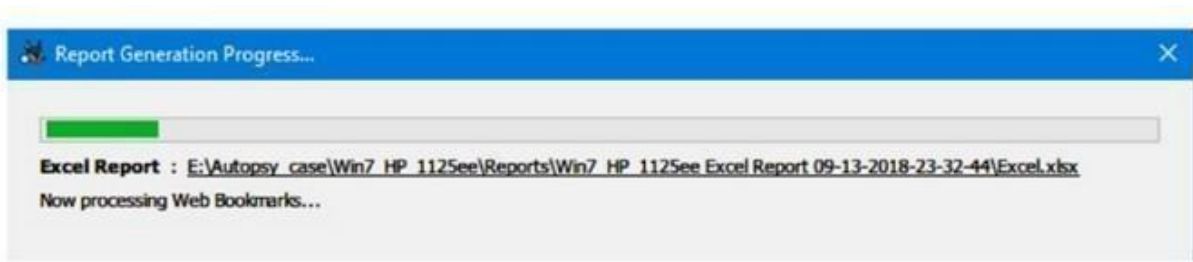


Figure 8.2 Report Generation Progress window

4. After it finishes generating the report, Autopsy will show you the link where your generated report is saved; click over this link to open the file using your default program (see Figure 8.3).
5. Finally, click “Close” to close the Report Generation Progress window.

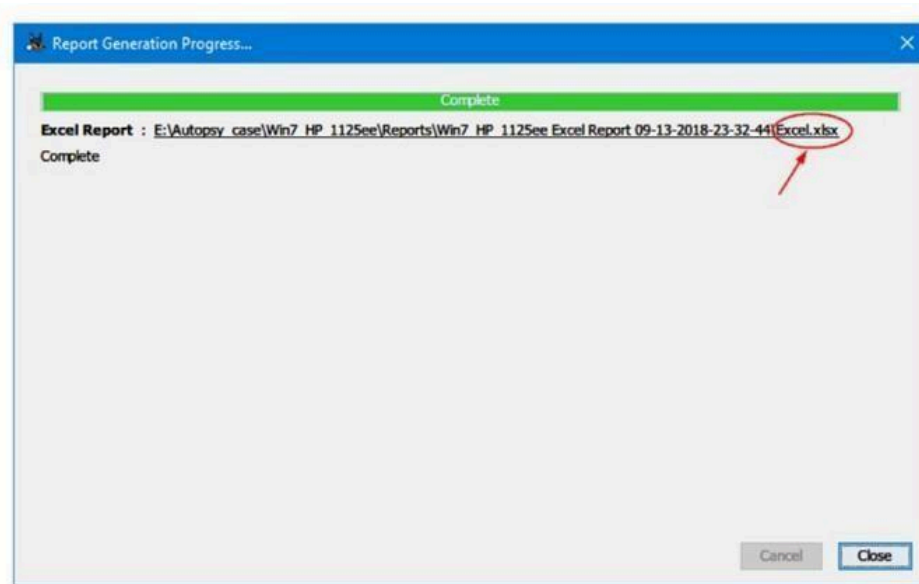


Figure 8.3. The location of Autopsy-generated report

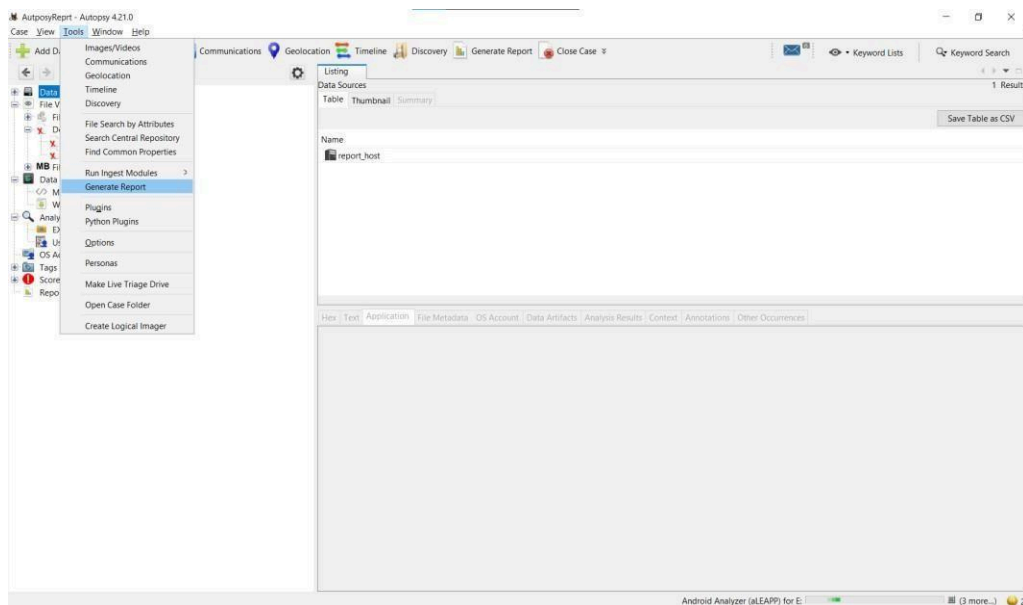
## RETRIEVING LAST SEVEN DAYS OF ACTIVITY USING AUTOPSY

Please note that as a part of Autopsy’s initial analysis, it will list the last seven days of activity— of web browsers (including web searches), installed programs, operating system, and recent changes to registry hives—of the supplied forensic image files in the Data Explorer panel under the “Extracted Content” section (see Figure 8.4). Remember that you need to activate the “Recent Activity” ingest module in order to retrieve this result.



Figure 8.4. Viewing the results of “Recent Activity” module: you must activate the “Recent Activity” ingest module to view this info.

## 8.7 Program and Output:



**Select and Configure Report Modules**

Report Modules:

- ☐ HTML Report
- ☒ Excel Report
- ☐ Files - Text
- ☐ Data Source Summary Report
- ☐ Save Tagged Hashes
- ☐ Extract Unique Words
- ☐ TSK Body File
- ☐ Google Earth KML
- ☐ CASE-UCO
- ☐ Portable Case

A report about results and tagged items in Excel (XLS) format.

*This report will be configured on the next screen.*

&lt; Back

Next &gt;

Finish

Cancel

Help

**Select which data source(s) to include**☒ E:

Uncheck All

Check All

&lt; Back

Next

Finish

Cancel

Help

Generate Report

×

Configure Report

Select which data to report on:

☒ All Results

☐ All Tagged Results

☐ Specific Tagged Results

Select All

Deselect All

Choose Result Types...

< Back

Next >

Finish

Cancel

Help

Report Generation Progress...

×

Complete

Excel Report : D:\sumeet \AutopsyReprt\Reports\AutopsyReprt Excel Report 10-02-2024-13-12-22\Excel.xlsx

Complete

Cancel

Close

|    | 1                               | 2             | 3             | 4        | 5                      | 6            | 7             | 8 | 9 | 10 |
|----|---------------------------------|---------------|---------------|----------|------------------------|--------------|---------------|---|---|----|
| 1  | Summary                         |               |               |          |                        |              |               |   |   |    |
| 2  |                                 |               |               |          |                        |              |               |   |   |    |
| 3  | Case Name:                      | AutopsyReprt  |               |          |                        |              |               |   |   |    |
| 4  | Case Number:                    | 002           |               |          |                        |              |               |   |   |    |
| 5  | Number of data sources in case: | 1             |               |          |                        |              |               |   |   |    |
| 6  |                                 |               |               |          |                        |              |               |   |   |    |
| 7  |                                 |               |               |          |                        |              |               |   |   |    |
| 8  |                                 |               |               |          |                        |              |               |   |   |    |
| 9  |                                 |               |               |          |                        |              |               |   |   |    |
| 10 |                                 |               |               |          |                        |              |               |   |   |    |
| 11 |                                 |               |               |          |                        |              |               |   |   |    |
| 12 |                                 |               |               |          |                        |              |               |   |   |    |
| 13 |                                 |               |               |          |                        |              |               |   |   |    |
| 14 |                                 |               |               |          |                        |              |               |   |   |    |
| 15 |                                 |               |               |          |                        |              |               |   |   |    |
| 16 |                                 |               |               |          |                        |              |               |   |   |    |
| 17 |                                 |               |               |          |                        |              |               |   |   |    |
| 18 |                                 |               |               |          |                        |              |               |   |   |    |
| 19 |                                 |               |               |          |                        |              |               |   |   |    |
| 20 |                                 |               |               |          |                        |              |               |   |   |    |
| 21 |                                 |               |               |          |                        |              |               |   |   |    |
| 22 |                                 |               |               |          |                        |              |               |   |   |    |
| 23 |                                 |               |               |          |                        |              |               |   |   |    |
| 24 |                                 |               |               |          |                        |              |               |   |   |    |
| 25 |                                 |               |               |          |                        |              |               |   |   |    |
| 26 |                                 |               |               |          |                        |              |               |   |   |    |
| 27 |                                 |               |               |          |                        |              |               |   |   |    |
| 28 |                                 |               |               |          |                        |              |               |   |   |    |
| 29 |                                 |               |               |          |                        |              |               |   |   |    |
| 30 |                                 |               |               |          |                        |              |               |   |   |    |
| 31 |                                 |               |               |          |                        |              |               |   |   |    |
|    | Summary                         | EXIF Metadata | Web Downloads | Metadata | User Content Suspected | Tagged Files | Tagged Rε ... | ⊕ |   |    |

## 8.8 Conclusion:

In conclusion, using Autopsy to generate a timeline report significantly enhances the efficiency and clarity of digital forensic investigations. The tool allows investigators to compile and visualize critical events in chronological order, aiding in the identification of patterns and correlations in user activity. This structured approach not only streamlines the analysis process but also facilitates effective communication of findings in legal contexts. Overall, Autopsy serves as a valuable resource in reconstructing events and supporting informed decision-making in forensic cases.

## 8.9 Review Questions based on Experiment:

1. What is the importance of the timeline report in the forensic investigation?

- The timeline report is vital in forensic investigations as it provides a chronological overview of events, helping to identify patterns and correlations in user activity. It allows investigators to reconstruct events and supports legal arguments by presenting a clear narrative of actions linked to specific timestamps. This clarity enhances the overall understanding of the case and aids in deriving meaningful conclusions from the evidence.



## 2. How to use Autopsy for the generation of timeline reports?

□ To generate timeline reports in Autopsy, first, create a case and add the data source (e.g., disk image) to it.

Navigate to the Timeline module under the "Analysis" tab, where you can specify the date range and types of events to include, such as file creation, access, or modification times. After setting your parameters, run the analysis, and Autopsy will compile the data into a timeline report. You can then export the report in various formats, such as CSV or HTML, for further review or presentation.

## 3. Give names of the similar software's used for report generation.

□ 1. FTK Imager

2. EnCase

3. Sleuth Kit

4. X1 social discovery

5. Cellebrite UFED